

| Runner     | HPC           | Container   | Status    | Time (s) |
|------------|---------------|-------------|-----------|----------|
| cwltool    | LUMI          | none        | ERR       | 87.0     |
| cwltool    | LUMI          | singularity | IN        | 79.0     |
| cwltool    | MareNostrum 5 | none        | <b>OK</b> | 8.0      |
| cwltool    | MareNostrum 5 | singularity | ERR       | 3.0      |
| cwltool    | Laptop        | none        | <b>OK</b> | 3.0      |
| cwltool    | Laptop        | docker      | <b>OK</b> | 5.0      |
| cwltool    | Laptop        | singularity | IN        | 4.0      |
| cwltool    | Cloud         | none        | <b>OK</b> | 5.0      |
| cwltool    | Cloud         | docker      | <b>OK</b> | 7.0      |
| cwltool    | Cloud         | singularity | IN        | 5.0      |
| streamflow | LUMI          | none        | ERR       | 113.0    |
| streamflow | LUMI          | singularity | TO        | 300.0    |
| streamflow | MareNostrum 5 | none        | ERR       | 16.0     |
| streamflow | MareNostrum 5 | singularity | ERR       | 1.0      |
| streamflow | Laptop        | none        | <b>OK</b> | 2.0      |
| streamflow | Laptop        | docker      | <b>OK</b> | 3.0      |
| streamflow | Laptop        | singularity | TO        | 31.0     |
| streamflow | Cloud         | none        | <b>OK</b> | 5.0      |
| streamflow | Cloud         | docker      | <b>OK</b> | 4.0      |
| streamflow | Cloud         | singularity | TO        | 30.0     |
| toil       | LUMI          | none        | ERR       | 124.0    |
| toil       | LUMI          | singularity | IN        | 113.0    |
| toil       | MareNostrum 5 | none        | ERR       | 18.0     |
| toil       | MareNostrum 5 | singularity | ERR       | 10.0     |
| toil       | Laptop        | none        | <b>OK</b> | 11.0     |
| toil       | Laptop        | docker      | <b>OK</b> | 18.0     |
| toil       | Laptop        | singularity | IN        | 14.0     |
| toil       | Cloud         | none        | <b>OK</b> | 15.0     |
| toil       | Cloud         | docker      | <b>OK</b> | 20.0     |
| toil       | Cloud         | singularity | IN        | 20.0     |

Table 1: Simple MPI workflow execution results across runners, platforms, and container backends with baseCommand: mpirun (TO: time-out, IN: invalid)

| Runner     | HPC           | Container   | Status    | Time (s) |
|------------|---------------|-------------|-----------|----------|
| cwltool    | LUMI          | none        | <b>OK</b> | 4.0      |
| cwltool    | LUMI          | singularity | ERR       | 5.0      |
| cwltool    | MareNostrum 5 | none        | <b>OK</b> | 3.0      |
| cwltool    | MareNostrum 5 | singularity | ERR       | 2.0      |
| cwltool    | Laptop        | none        | <b>OK</b> | 3.0      |
| cwltool    | Laptop        | docker      | <b>OK</b> | 4.0      |
| cwltool    | Laptop        | singularity | ERR       | 3.0      |
| cwltool    | Cloud         | none        | <b>OK</b> | 4.0      |
| cwltool    | Cloud         | docker      | ERR       | 5.0      |
| cwltool    | Cloud         | singularity | ERR       | 6.0      |
| streamflow | LUMI          | none        | ERR       | 3.0      |
| streamflow | LUMI          | singularity | ERR       | 3.0      |
| streamflow | MareNostrum 5 | none        | ERR       | 1.0      |
| streamflow | MareNostrum 5 | singularity | ERR       | 1.0      |
| streamflow | Laptop        | none        | ERR       | 2.0      |
| streamflow | Laptop        | docker      | ERR       | 1.0      |
| streamflow | Laptop        | singularity | ERR       | 1.0      |
| streamflow | Cloud         | none        | ERR       | 2.0      |
| streamflow | Cloud         | docker      | ERR       | 3.0      |
| streamflow | Cloud         | singularity | ERR       | 2.0      |
| toil       | LUMI          | none        | <b>OK</b> | 14.0     |
| toil       | LUMI          | singularity | ERR       | 20.0     |
| toil       | MareNostrum 5 | none        | ERR       | 8.0      |
| toil       | MareNostrum 5 | singularity | ERR       | 9.0      |
| toil       | Laptop        | none        | <b>OK</b> | 11.0     |
| toil       | Laptop        | docker      | ERR       | 4.0      |
| toil       | Laptop        | singularity | ERR       | 14.0     |
| toil       | Cloud         | none        | <b>OK</b> | 14.0     |
| toil       | Cloud         | docker      | ERR       | 2.0      |
| toil       | Cloud         | singularity | ERR       | 19.0     |

Table 2: Simple MPI workflow execution results across runners, platforms, and container backends with MPIRequirement (TO: time-out, IN: invalid)